

COM668 Project

Requirements and User Stories

Requirements and User Stories

- Requirements engineering
- Story driven approaches
- The best approach

Range of approaches exist

- Wide range of approaches exist
- Some fit better with specific methodologies
- Others are a better fit to personal skills, ability, perspective
- We will consider two different approaches
- Research further and pick an approach that suits your project
- Justify your decisions in your reports

Requirements engineering

A traditional perspective

Requirements engineering focus

- Define requirements of the end product at the start of the project
- Thorough, albeit for a brief moment in the projects life
- Limited interest in the broader methodology
- Traditionally no explicit change management process

Approaches

- There are numerous approaches and tools
- The Volere template is quite popular
 - Fairly easy to understand
 - Relatively lightweight to implement
 - Based around a template (free for academic use)
 - Guides defining requirements and context they reside within

Volere

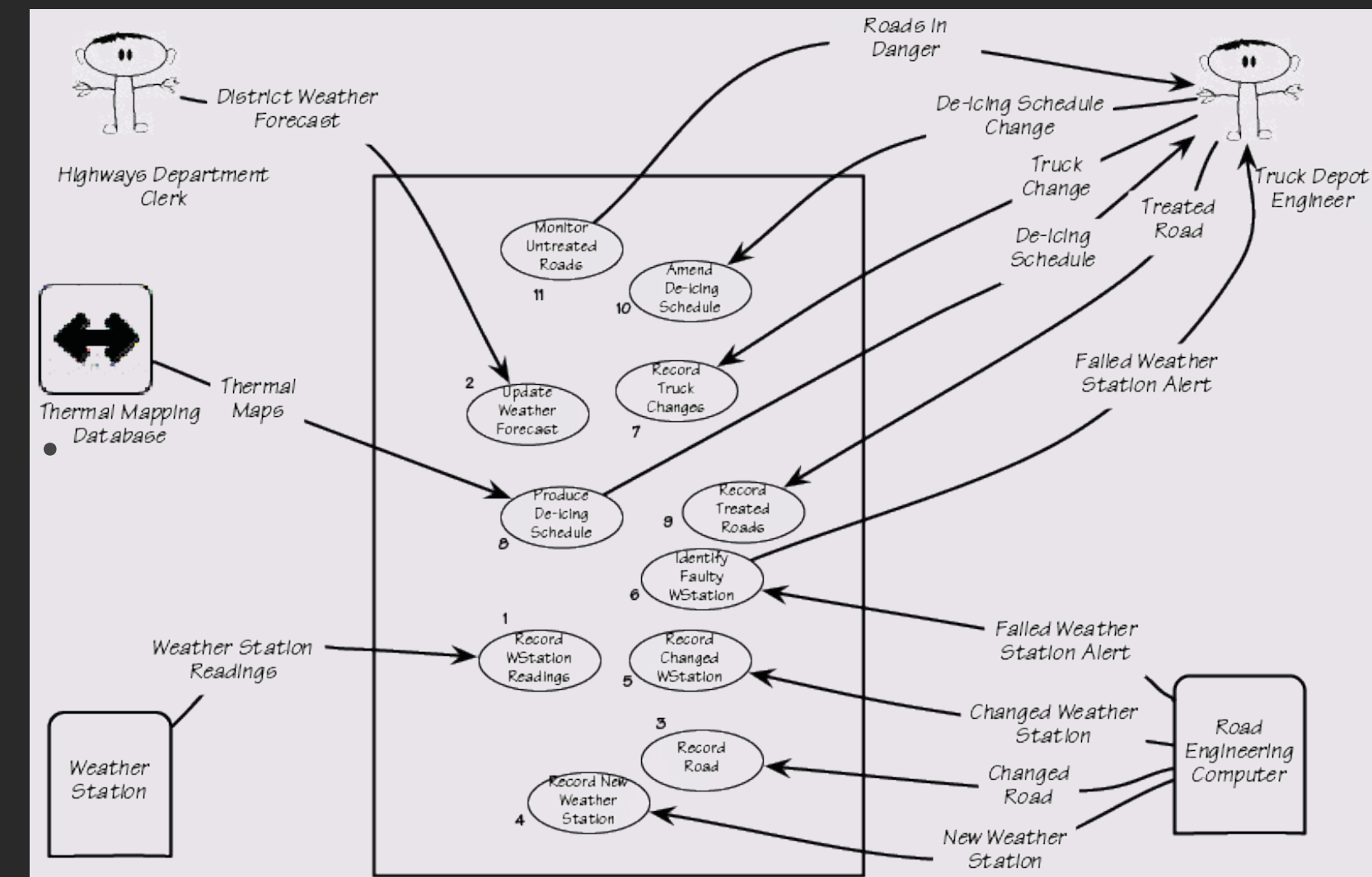
To wish or to want

Volere overview

- This is a very brief overview, you should have a closer look
- Why consider such a traditional approach?
 - Traditional is not necessarily the same as flawed or outdated
 - Structure can help when inexperienced
- Download the template and read the guidance
 - Do not pay, it is free to students
 - Download from: *<http://www.volere.co.uk>*
- Going to focus on just a few key parts today

Scope of Product

- Strives to identify what must be done
 - Similar to creating initial PBI's, or Epics, but the perspective is different
- Uses tools such as Use Case Diagrams
 - Inspect each business Use Case with stakeholders
 - Identify boundaries between users and system
 - Identify who or what does what
 - Relate items to Business Use Cases
- Strong points
 - Includes stakeholders
 - Business value is important
 - Tries to identify tasks
- Weak points
 - This is almost always an upfront once only process
 - The whole development team is not necessarily involved



Functional Requirements

- Fundamental to the Volere Requirements Specification Template
- Functional requirement
 - Atomic functional element of the product
 - Arrived at through a process of requirement decomposition
 - Together they defines the required functionality of the product
- Function requirements should be
 - Measurable
 - Testable
 - Traceable
 - Detailed
- Specified in detail using a Volere Snow Card

Volere Snow Card

Requirement #:

Requirement Type:

Event/BUC/PUC #:

Description:

Rationale:

Originator:

Fit Criterion:

Customer Satisfaction:

Customer Dissatisfaction:

Priority:

Conflicts:

Supporting Materials:

History:

Volere

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Volere Snow Card

Requirement #:

Requirement Type:

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Description:

Rationale:

Originator:

Fit Criterion:

Customer Satisfaction:

Priority:

Supporting Materials:

History:

Customer Dissatisfaction:

Conflicts:

Unique identifier. Which helps to make reference to the requirement during the later stages of development.

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Requirement #:

Requirement Type:

Event/BUC/PUC #:

Description:

Rationale:

Originator:

Fit Criterion:

What the requirement is about. As a one sentence description, from the perspective of a stakeholder.

Customer Satisfaction:

Customer Dissatisfaction:

Priority:

Conflicts:

Supporting Materials:

History:

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Requirement #:

Requirement Type:

Event/BUC/PUC #:

Description:

Rationale:

Originator:

Fit Criterion:

Customer Satisfaction:

Priority:

Supporting Materials:

History:

Volere requirement type,
e.g. Functional, Non-Functional etc.

Customer Dissatisfaction:

Conflicts:

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Requirement #:

Requirement Type:

Event/BUC/PUC #:

Description:

Rationale:

Originator:

Fit Criterion:

A brief statement that justifies why the requirement is important. Help to make a case for implementing it.

Customer Satisfaction:

Customer Dissatisfaction:

Priority:

Conflicts:

Supporting Materials:

History:

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Requirement #:

Requirement Type:

Event/BUC/PUC #:

Description:

Rationale:

An objective measure that is used later to test if the solution meets this requirement. Similar to a definition of done from Agile.

Originator:

Fit Criterion:

Customer Satisfaction:

Customer Dissatisfaction:

Priority:

Conflicts:

Supporting Materials:

History:

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Requirement #:

Requirement Type:

Event/BUC/PUC #:

Description:

Rationale:

Originator:

Fit Criterion:

Lists the Requirement # of any requirements that this requirement conflicts with .
Useful during planning.

Customer:

Dissatisfaction:

Priority:

Conflicts:

Supporting Materials:

History:

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Requirement #:

Requirement Type:

Event/BUC/PUC #:

Description:

Rationale:

Originator:

Fit Criterion:

Customer Satisfaction:

Customer Dissatisfaction:

Priority:

Conflicts:

Supporting Materials:

History:

Volere

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Requirements Specification

- That is a lot to define
- Thorough but...
 - What if something changes?
 - What if something is not possible?
- Criticism
 - This is an upfront time cost, usually based on assumptions
 - A risk?
- Counter argument
 - Approaches such as Agile introduce risk through uncertainty
 - How can you start without knowing the end product

Story driven approaches

Agile and beyond

INVEST

- Agile recommends that Product Backlog Items should be:
 - Independent: Business value over technical efficiencies
 - Negotiable: by business stakeholders and development team
 - Valuable: Customer centred, which brings clear business value
 - Estimable: Understood enough to agree how long it will take
 - Small: 1 or 2 team days to complete, max 1/4 of a sprint
 - Testable: Clear criteria for when '*done*' agreed by PO and team
- However, first need to develop the overarching user needs
- Story driven methods can be useful for this
 - And are not only of value when taking an Agile approach

User Stories

- Title: [One line describing an activity]
- Narrative: As a role I want to be able to feature so that benefit
- Acceptance criteria: [Presented as scenarios]
- Scenario structure: Title [that is clearly different from other scenarios]
 - Given context
 - And some more context...
 - When event
 - Then outcome
 - And another outcome...

ATM user story example

- Title: Account Holder withdraws cash
- Narrative: As an account holder I want to be able to withdraw cash from an ATM so that I can get money when the bank is closed
- Acceptance criteria:

ATM user story - Scenario 1

- Title: Account Holder withdraws cash
- Narrative: As an account holder I want to be able to withdraw cash from an ATM so that I can get money when the bank is closed
- Acceptance criteria:
- Scenario 1: Account has sufficient funds
 - Given the account balance is sufficient for the amount requested
 - And the bank card is valid
 - And the ATM has sufficient funds
 - When the account holder requests a cash withdrawal
 - Then the ATM should dispense the requested amount
 - And the account balance should be debited by the dispensed amount
 - And the bank card should be returned

ATM user story - Scenario 2

- Title: Account Holder withdraws cash
- Narrative: As an account holder I want to be able to withdraw cash from an ATM so that I can get money when the bank is closed
- Acceptance criteria:
- Scenario 2: Account has insufficient funds
 - Given the account balance is insufficient for the amount requested
 - And the bank card is valid
 - And the ATM has sufficient funds
 - When the account holder requests a cash withdrawal
 - Then the ATM should inform the customer they have insufficient funds
 - And ensure that no cash is dispensed
 - And ensure that the account balance is not adjusted
 - And the bank card should be returned

The best approach

Discuss this with your Mentor

The best approach

- Ultimately need to think of the bigger picture
 - User Centred Design is an important design methodology
 - Story driven approaches are a more natural fit to this
 - However, you are largely your user in this project
- Volere, or a similar engineering approach adds structure
 - You might find this helpful given you lack of experience
 - It might be more efficient in the absence of third-party users
- Any technique used must be
 - Justified in its selection
 - Rigorous and detailed in its delivery
- Research and the deep dive lecture will give you a fuller insight
- Discuss the options and opinions with your Mentor

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